

# Compact Molecular Descriptors for Honey Bee Toxicity Screening: A Structure-Aware Comparison of Classical and Quantum Kernels

Anonymous Authors

**Abstract**—Species-specific pesticide toxicity data are limited, which motivates compact and reproducible screening models. We evaluate ten physicochemical and structural molecular descriptors (X10) for binary acute toxicity to the honey bee, *Apis mellifera*. The curated domain contains 893 molecules, with 712 development compounds assigned to five frozen structure-aware folds and a cluster-disjoint 181-molecule historical holdout. Class-weighted logistic regression, random forest, and radial basis function (RBF) support vector machines are compared with exact-statevector product and entangled IQP-ZZ fidelity kernels on identical descriptors and molecules. Random forest gives the highest development ranking performance (AUROC 0.756, 95% bootstrap interval 0.714–0.793), while the matched RBF reaches 0.710 on the historical holdout. Removing the explicit organophosphorus count lowers development AUROC by 0.048 for logistic regression and 0.052 for random forest. Removing both partition descriptors lowers RBF AUROC by 0.047. The IQP-ZZ kernel improves development AUROC over matched RBF by 0.018 (0.004–0.032), but not historical-holdout AUROC (−0.006, −0.026–0.015). Quantum and RBF kernels are strongly aligned and disagree on few classifications. X10 therefore contains useful screening signal, but the experiment does not establish a generalizable quantum advantage.

**Index Terms**—honey bee toxicity, molecular descriptors, pesticide screening, quantum kernel, support vector machine, structure-aware validation

## I. INTRODUCTION

Honey bees are economically and ecologically important pollinators, yet species-specific toxicity measurements cover only a small fraction of the chemical space relevant to agriculture. Public ecotoxicology resources are large but require structural standardization, deduplication, endpoint selection, and label harmonization before supervised learning [1]. ApisTox addressed part of this gap by curating a benchmark for pesticide and agrochemical toxicity to *Apis mellifera* [2].

Recent evaluation on ApisTox found that molecular fingerprints and graph kernels can outperform more complex graph neural and pretrained molecular models [3]. This result motivates a complementary question: how much predictive information is retained by a much smaller, interpretable descriptor panel? A compact representation is attractive for auditing, controlled ablation, and near-term quantum kernel experiments, where qubit and simulation costs grow with input dimension.

This work contributes: 1) a leakage-controlled evaluation of ten compact descriptors under frozen, cluster-disjoint folds; 2) pre-specified ablations of an organophosphorus motif and two partition descriptors; 3) a matched classical-quantum comparison using exact fidelity kernels; and 4) a hash-addressed workflow connecting data, predictions, figures, and the manuscript. The goal is not to claim quantum advantage, but to test whether a quantum feature map induces behavior that is detectably different from an RBF kernel under equal molecular coordinates.

## II. BACKGROUND

### A. Bee Toxicity Prediction

ApisTox contains standardized structures, toxicity labels, and challenging predefined splits [2]. Fingerprints such as ECFP encode local molecular neighborhoods [4], while graph kernels compare molecular topology directly [5]. Both are strong references for molecular property prediction. The present study instead uses a curated subset for which ten conventional and in-house descriptors are finite and available. It is therefore a targeted follow-up, not a reproduction of the full ApisTox benchmark.

### B. Quantum Kernels

Quantum feature maps encode classical inputs into states in a Hilbert space; state overlap then defines a kernel for a conventional classifier [6], [7]. This formulation permits a controlled comparison because the optimization after kernel construction is the same support vector machine used classically. It also requires caution: different kernels can be classically simulated, and favorable validation performance alone is not evidence of computational advantage.

## III. MATERIALS AND METHODS

### A. Curated Domain and Endpoint

The binary endpoint uses the strongest available acute LD50 per molecule: toxic when  $LD50 \leq 11 \mu\text{g}/\text{bee}$  and non-toxic otherwise. The curated domain contains organic, metal-free molecules with finite X10 values. Development and holdout labels are moderately imbalanced (Table I). The holdout was inspected during descriptor pre-screening before this study;

TABLE I  
CURATED MOLECULAR DOMAIN AND FIXED EVALUATION PROTOCOL.

| Property                       | Value     |
|--------------------------------|-----------|
| All curated molecules          | 893       |
| Development molecules          | 712       |
| Historical holdout molecules   | 181       |
| Development toxic / non-toxic  | 222 / 490 |
| Holdout toxic / non-toxic      | 49 / 132  |
| Development / holdout clusters | 374 / 88  |
| Frozen development folds       | 5         |
| Molecular descriptors          | 10        |

we therefore call it a *historical holdout*, not an untouched confirmatory test.

The split was constructed using 1024-bit Morgan fingerprints with radius two, chirality enabled, Butina clustering at cutoff 0.60 [8], and group-aware stratification. Development and holdout share no Butina cluster. Each development cluster belongs to exactly one of five frozen folds.

### B. X10 Representation and Hypotheses

X10 includes molecular weight (MolWt), lipophilicity (MolLogP), topological polar surface area with sulfur and phosphorus contributions (TPSA), hydrogen bond donors, rotatable bonds, aromatic rings, halogen count, an explicit organophosphorus motif count ( $n_{OP}$ ), an in-house hexadecane/water partition estimate (LiPHEX), and an in-house polar solvent-accessible surface fraction. The fixed feature order is part of the run hash.

We test five hypotheses: X10 separates the toxicity classes;  $n_{OP}$  adds signal beyond global physicochemical variables; MolLogP and LiPHEX contribute complementary partition information; quantum and RBF kernels differ under matched coordinates; and predictive ranking persists on cluster-disjoint molecules. Ablations remove  $n_{OP}$ , MolLogP, LiPHEX, or both partition descriptors without changing any other protocol component.

### C. Classical Models

The classical models are class-weighted logistic regression, RBF-SVC, and a 300-tree random forest. Standardization is fitted only on rows available to a training partition. Each frozen fold serves once as outer validation. Within each outer-training partition, four-fold stratified group cross-validation, grouped by Butina cluster, selects hyperparameters by AUROC. Logistic  $C$  is chosen from  $\{0.01, 0.1, 1, 10, 100\}$ ; RBF uses  $C \in \{0.1, 1, 10, 100\}$  and  $\gamma \in \{0.01, 0.03, 0.1, 0.3, \text{scale}\}$ . Random forest depth and minimum leaf size are selected from a small predeclared grid. The five outer predictions are pooled so that each development molecule contributes exactly one out-of-fold (OOF) score.

### D. Matched Simulated Quantum Kernels

One standardized descriptor is assigned to each of ten qubits. The product control map is

$$|\phi_P(\mathbf{x})\rangle = \bigotimes_{j=1}^{10} R_y(sx_j)|0\rangle. \quad (1)$$

The entangled IQP-ZZ map starts in a uniform superposition and assigns basis state  $\mathbf{z} \in \{-1, 1\}^{10}$  the phase

$$\exp\left[i\left(s\sum_j x_j z_j + \eta s^2 \sum_{j=1}^9 x_j x_{j+1} z_j z_{j+1}\right)\right], \quad (2)$$

with  $\eta = 1$ . Exact complex statevectors define  $K_Q(\mathbf{x}, \mathbf{x}') = |\langle\phi(\mathbf{x})|\phi(\mathbf{x}')\rangle|^2$ . No sampling noise or hardware backend is used.

The angle scale  $s \in \{0.125, 0.25, 0.5, 1\}$  and  $C \in \{0.1, 1, 10, 100\}$  are selected using the same nested folds and criterion as RBF. An RBF kernel with four  $\gamma$  values and the same four  $C$  values is recomputed in the quantum pipeline, giving each kernel family a 16-point search budget. Every Gram matrix is checked for finiteness, symmetry, unit diagonal, and positive semidefiniteness.

### E. Metrics and Statistical Analysis

AUROC is primary; AUPRC, balanced accuracy, and Matthews correlation coefficient (MCC) are secondary. We use 2,000 class-stratified bootstrap resamples for 95% intervals. Model and ablation differences use paired resampling of molecule IDs. Kernel geometry is characterized by centered alignment, target alignment, and entropy-based effective rank. To probe domain shift, holdout molecules are divided into tertiles by nearest Euclidean distance in standardized X10 space. All random seeds, folds, configurations, predictions, and output files are recorded with SHA-256 hashes.

## IV. RESULTS

### A. Compact Descriptors Carry Predictive Signal

All X10 models rank toxic molecules above chance in development and in the historical holdout (Table II). Random forest is strongest in development at AUROC 0.756, while the matched RBF is strongest on the historical holdout at 0.710. The larger holdout intervals reflect its smaller sample size (181 molecules, 49 positive). Threshold metrics do not follow the same ordering: RBF has the highest development MCC (0.466), whereas logistic regression has the highest X10 holdout MCC (0.398). This supports the X10 signal hypothesis while showing that ranking and fixed-threshold behavior must be reported separately.

### B. Structural and Partition Ablations

Removing  $n_{OP}$  decreases development AUROC by 0.048 for logistic regression (95% paired interval 0.021–0.073) and 0.052 for random forest (0.028–0.076), but only 0.002 for RBF (−0.028–0.031). Thus the explicit motif adds measurable signal, although its effect depends on model family. Only 12

TABLE II  
X10 PREDICTIVE PERFORMANCE. PARENTHESES SHOW 95% STRATIFIED BOOTSTRAP INTERVALS FOR AUROC.

| Evaluation         | Model                | AUROC (95% CI)      | AUPRC | Balanced Acc. | MCC   |
|--------------------|----------------------|---------------------|-------|---------------|-------|
| Development OOF    | Random forest        | 0.756 (0.714–0.793) | 0.671 | 0.681         | 0.419 |
| Development OOF    | Logistic regression  | 0.717 (0.671–0.761) | 0.629 | 0.680         | 0.377 |
| Development OOF    | Quantum IQP-ZZ map   | 0.732 (0.687–0.774) | 0.636 | 0.688         | 0.458 |
| Development OOF    | Quantum product map  | 0.729 (0.686–0.770) | 0.618 | 0.684         | 0.450 |
| Development OOF    | RBF kernel (matched) | 0.714 (0.670–0.757) | 0.628 | 0.692         | 0.466 |
| Historical holdout | Logistic regression  | 0.689 (0.589–0.786) | 0.516 | 0.693         | 0.398 |
| Historical holdout | Random forest        | 0.689 (0.597–0.783) | 0.518 | 0.645         | 0.337 |
| Historical holdout | RBF kernel (matched) | 0.710 (0.622–0.795) | 0.543 | 0.611         | 0.282 |
| Historical holdout | Quantum IQP-ZZ map   | 0.704 (0.614–0.789) | 0.505 | 0.588         | 0.245 |
| Historical holdout | Quantum product map  | 0.703 (0.603–0.790) | 0.487 | 0.595         | 0.253 |

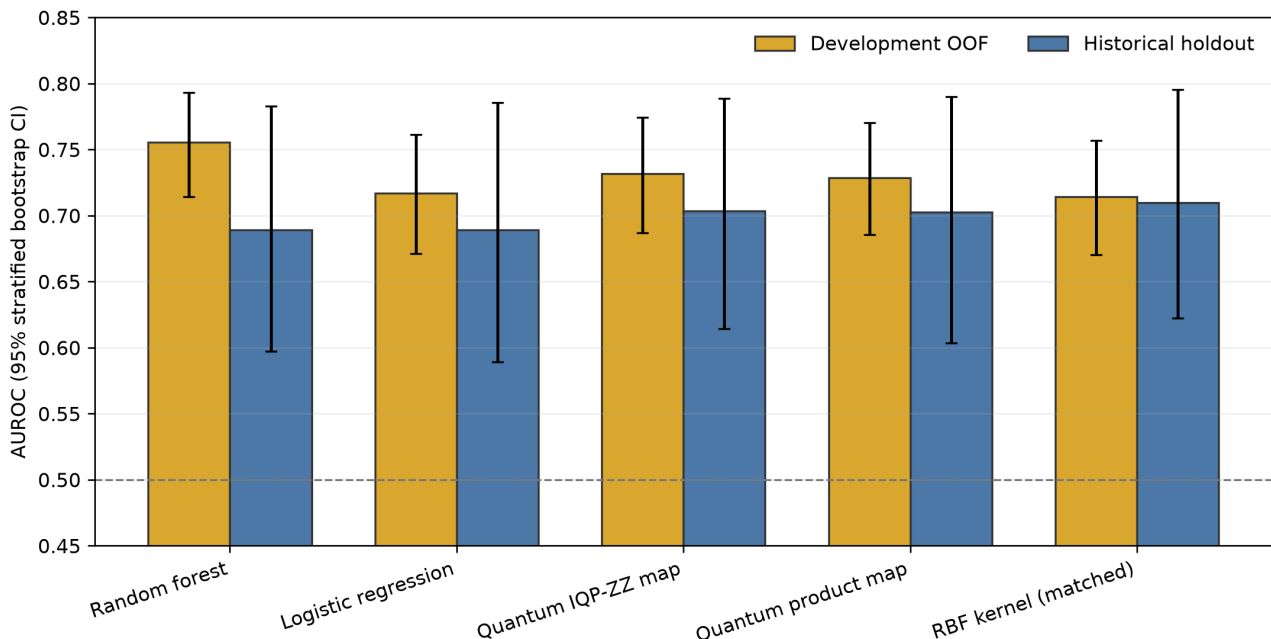


Fig. 1. X10 AUROC with 95% class-stratified bootstrap intervals. The historical holdout was previously inspected and is not a confirmatory test.

holdout molecules have  $n_{OP} > 0$ , so motif-specific holdout estimates are too sparse for a stable conclusion.

MolLogP and LiPHEX have Spearman correlation 0.405 in development. For RBF, removing MolLogP lowers AUROC by 0.024 (0.004–0.044), removing LiPHEX lowers it by 0.031 (0.002–0.062), and removing both lowers it by 0.047 (0.013–0.079). This supports complementary partition information for the RBF geometry. LiPHEX removal changes logistic and random-forest AUROC by only -0.003 and -0.004, respectively, with intervals crossing zero (Table III).

### C. Classical Versus Quantum Kernels

In development, the product and IQP-ZZ kernels reach AUROC 0.729 and 0.732, respectively, compared with 0.714 for matched RBF. Paired improvements are 0.015 (0.001–0.030) and 0.018 (0.004–0.032). These gains do not transfer to the historical holdout: product minus RBF is -0.007 (-0.038–0.024), and IQP-ZZ minus RBF is -0.006 (-0.026–0.015).

TABLE III  
DEVELOPMENT AUROC CHANGE  $\Delta$  = X10 MINUS ABLATED REPRESENTATION.

| Removed        | LR     | RBF    | RF     |
|----------------|--------|--------|--------|
| $n_{OP}$       | +0.048 | +0.002 | +0.052 |
| MolLogP        | +0.006 | +0.024 | +0.011 |
| LiPHEX         | -0.003 | +0.031 | -0.004 |
| Both partition | +0.012 | +0.047 | +0.009 |

The kernels are also less behaviorally distinct than their names suggest. Across outer-training folds, centered alignment with RBF is 0.989 for the product map and 0.992 for IQP-ZZ (Table IV). Their OOF class decisions differ from RBF on only 1.4% and 1.1% of development molecules; holdout disagreement is 1.7% and 2.8%. IQP-ZZ has greater effective rank than RBF but remains strongly aligned. The experiment

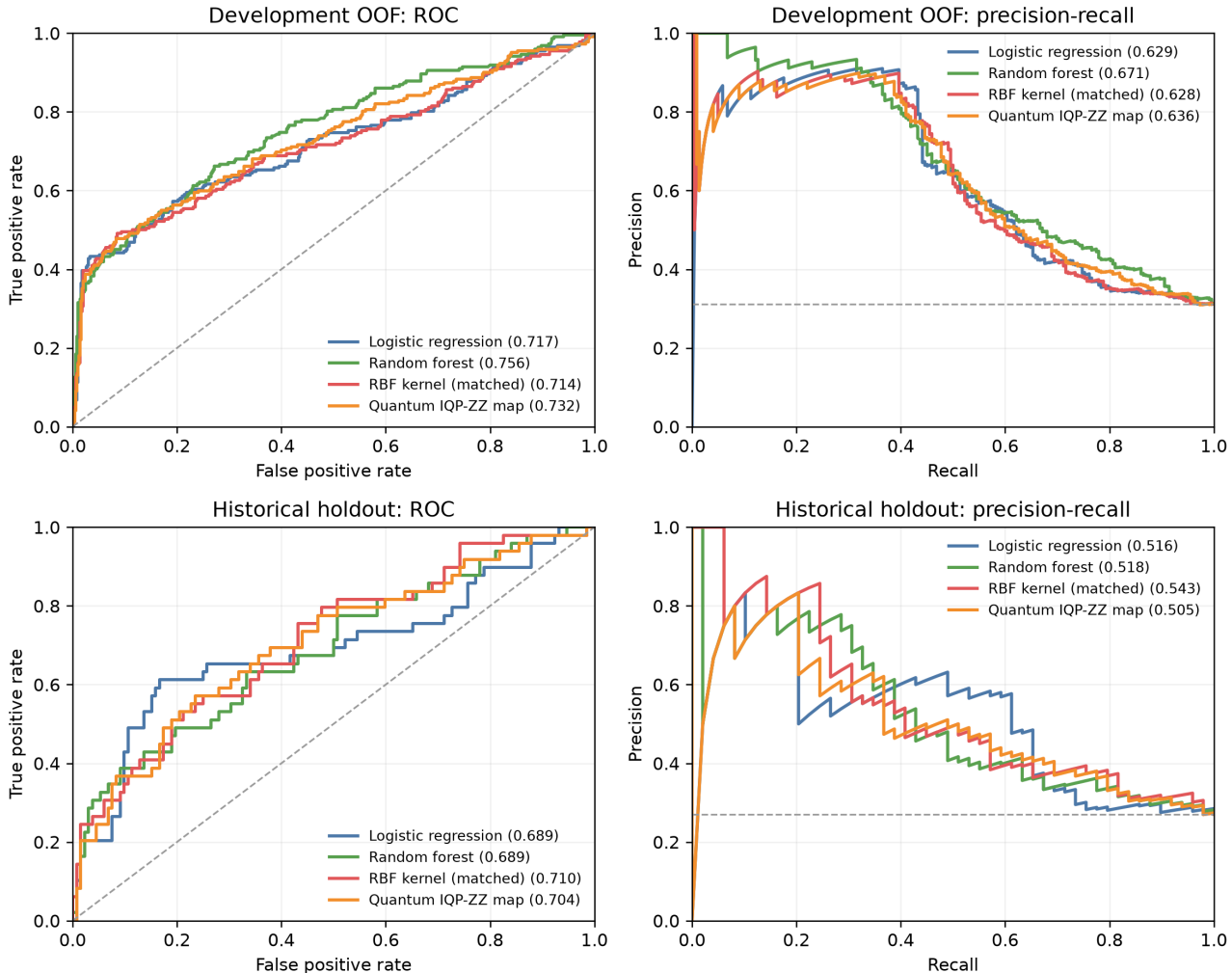


Fig. 2. ROC and precision-recall curves for selected X10 models. Legend values are AUROC (left) and AUPRC (right). Dashed precision-recall baselines show the positive-class prevalence.

TABLE IV  
X10 KERNEL GEOMETRY ACROSS OUTER-TRAINING FOLDS (MEAN  $\pm$  SD).

| Kernel  | Eff. rank       | Target align.     | RBF align.        |
|---------|-----------------|-------------------|-------------------|
| Product | 1.82 $\pm$ 0.70 | 0.090 $\pm$ 0.009 | 0.989 $\pm$ 0.004 |
| IQP-ZZ  | 3.15 $\pm$ 0.02 | 0.104 $\pm$ 0.012 | 0.992 $\pm$ 0.001 |
| RBF     | 2.28 $\pm$ 0.01 | 0.097 $\pm$ 0.011 | 1.000 $\pm$ 0.000 |

therefore detects a small validation ranking difference, not a robust quantum advantage.

#### D. Structure-Disjoint Generalization

X10 AUROC remains between 0.689 and 0.710 on the historical holdout, providing evidence that ranking signal persists across disjoint Butina clusters. The signal weakens with descriptor-space distance: near-tertile AUROC is 0.829 for random forest and 0.760–0.776 for the three kernel models, whereas far-tertile AUROC falls to 0.601–0.642. Error rates rise from 0.15–0.20 in the near tertile to 0.32–0.35 in the

far tertile (Fig. 4). This pattern identifies an applicability limitation shared by classical and quantum methods.

## V. DISCUSSION

The results support the principal claim that a ten-descriptor representation contains useful acute bee-toxicity signal under structure-aware evaluation. The strongest development model is not the strongest holdout model, and model-specific ablations matter:  $n_{OP}$  is important for linear and tree models, while both partition descriptors jointly shape the RBF kernel. These findings favor retaining X10 as the common comparison space rather than selecting a smaller panel from the historical holdout.

The quantum comparison is intentionally conservative. Both maps use the same standardized coordinates, molecules, folds, class weights, and search budget as RBF. Exact statevectors remove shot noise and backend drift, isolating the feature-map question. IQP-ZZ yields a statistically positive OOF AUROC difference, but the difference reverses slightly and becomes un-

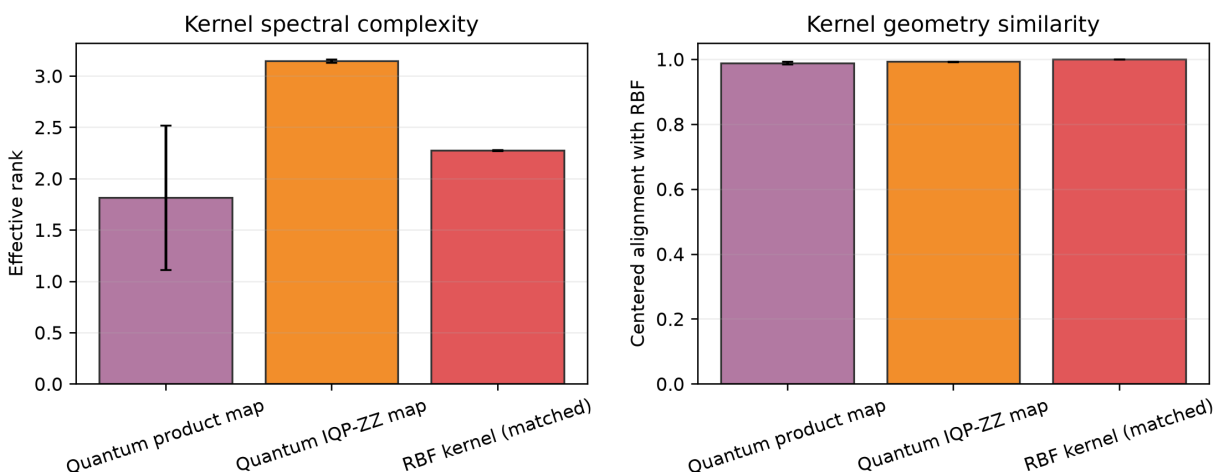


Fig. 3. X10 kernel geometry across outer-training folds. Error bars are one standard deviation. Both quantum kernels remain strongly aligned with the matched RBF despite different effective ranks.

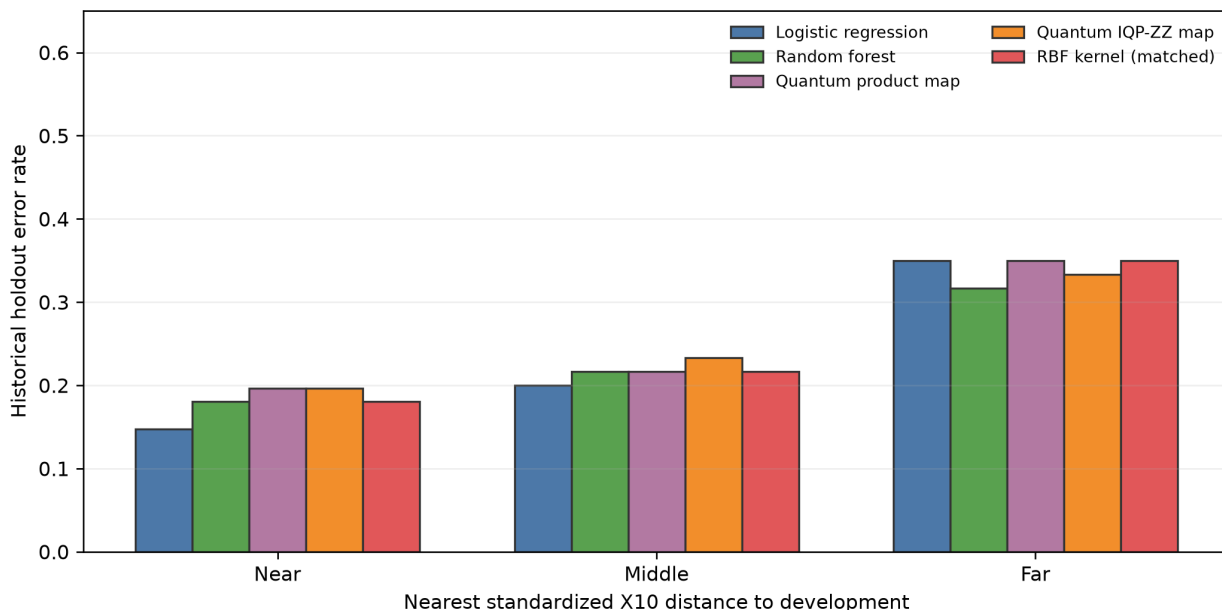


Fig. 4. Historical-holdout error rate by nearest standardized X10 distance to development. Each tertile contains approximately 60 molecules.

certain on the holdout. High kernel alignment and low decision disagreement further indicate that the tested quantum maps mostly reproduce an RBF-like neighborhood structure. This is useful negative evidence: a different circuit construction or a representation containing richer structural coordinates would be needed to test genuinely distinct geometry.

Several limitations bound the claims. First, the historical holdout was consulted during earlier descriptor pre-screening and cannot support a confirmatory claim. Second, X10 is available for 893 curated molecules rather than the full ApisTox collection, and two in-house descriptors are not yet recomputable from public code. Third, the endpoint collapses exposure routes into one thresholded label. Fourth, exact ten-

qubit simulation demonstrates a quantum kernel method, not computational speedup or hardware robustness. Finally, the distance analysis uses standardized descriptors; Butina cluster separation supplies the structural control, but a future regional set should also report fingerprint similarity and independent toxicity provenance.

## VI. REPRODUCIBILITY

The workflow is split into data audit, classical models, quantum kernels, statistical analysis, and manuscript generation. Source CSVs remain outside version control. A canonical split hash fixes molecule IDs, labels, clusters, and folds. Each run stores package versions, Git state, configurations, per-fold parameters, OOF and holdout predictions, metrics, and output

hashes. The notebook loads committed result bundles when expensive experiments are not recomputed and also provides explicit execution cells for every phase.

## VII. CONCLUSION

Ten compact descriptors distinguish toxic from non-toxic pesticides under cluster-disjoint evaluation. Explicit organophosphorus structure and two partition descriptors contribute model-dependent information. Simulated quantum kernels produce small OOF ranking gains and measurably different spectra, but their strong alignment with RBF and lack of holdout improvement do not support a generalizable quantum advantage. The resulting reproducible baseline is suitable for future circuit studies and independently labeled regional validation.

## REFERENCES

- [1] J. H. Olker, C. M. Elonen, A. Pilli, A. Anderson, B. Kinziger *et al.*, “The ECOTOXicology knowledgebase: A curated database of ecologically relevant toxicity tests to support environmental research and risk assessment,” *Environmental Toxicology and Chemistry*, vol. 41, no. 6, pp. 1520–1539, 2022.
- [2] J. Adamczyk, J. Poziemski, and P. Siedlecki, “ApisTox: A new benchmark dataset for the classification of small molecules toxicity on honey bees,” *Scientific Data*, vol. 12, no. 1, p. 5, 2025.
- [3] —, “Evaluating machine learning models for predicting pesticides toxicity to honey bees,” *arXiv preprint arXiv:2503.24305*, 2025.
- [4] D. Rogers and M. Hahn, “Extended-connectivity fingerprints,” *Journal of Chemical Information and Modeling*, vol. 50, no. 5, pp. 742–754, 2010.
- [5] S. V. N. Vishwanathan, N. N. Schraudolph, R. Kondor, and K. M. Borgwardt, “Graph kernels,” *Journal of Machine Learning Research*, vol. 11, pp. 1201–1242, 2010.
- [6] V. Havlicek, A. D. Corcoles, K. Temme *et al.*, “Supervised learning with quantum-enhanced feature spaces,” *Nature*, vol. 567, pp. 209–212, 2019.
- [7] M. Schuld, “Supervised quantum machine learning models are kernel methods,” *arXiv preprint arXiv:2101.11020*, 2021.
- [8] D. Butina, “Unsupervised data base clustering based on daylight’s fingerprint and tanimoto similarity: A fast and automated way to cluster small and large data sets,” *Journal of Chemical Information and Computer Sciences*, vol. 39, no. 4, pp. 747–750, 1999.